How Does Linguistic Knowledge Enter AI?

[MANOVA AI]
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Structured Resources and the Future of Linguistics

I will start with a clarification. Language information is not the same as linguistic information. The former is the means necessary for communication; the latter is information specific to the scientific field of linguistics, which studies language in all its forms and facets.

Linguistic information enters AI in the same way as information about all other scientific fields and spheres of life does. The AI model is first equipped with a method of tokenization, which is part of the language information of the model. Tokenization is a method for parsing the linear flow of language into small pieces called tokens that are not associated with meaning. In other words, tokens are not linguistic units. For ease of perception, tokens can be seen as frequent sequences of characters.

Linguistic information comes from linguistic books, articles, and other resources—not necessarily only written ones—that have been used for AI training. Thus, linguistics does not have a special status in an AI model but is treated like all other fields of knowledge.

Since linguistics investigates language as a phenomenon, many linguists seem to believe that all linguistic resources are very helpful for AI training. But is this so?

Given that AI operates exclusively with linear sequences of tokens, some resources are less suitable than others for training. For example, structured resources such as (in alphabetical order) corpora, grammaticons, lexicons, and the like require special effort on the computer scientists' side before they can be used as training data. Structured resources first have to be linearized and cleansed of any metalanguage because annotation labels in an annotated text are too numerous and will certainly be reflected in the embeddings, in which the combinability of a token is accounted for.

Although any sequence of characters can be tokenized and used for training, the decision about which data are appropriate for training must be made by a human; that is, the selection of the training data requires qualified human effort and is costly. However, the linguistic community is relatively small, which means that there is little incentive. Of course, a linguist could ask: Why do structured resources require additional effort to be used for AI training if they contain correct grammatical information? Some linguists even refer to such resources as the gold standard.

Besides the aforementioned issues of the non-linear organization of text and excessive amounts of metalanguage, the data should reflect natural human discourse, since the aim of an AI model is to model human language use. This is, as a rule, not the case with structured linguistic resources. Who speaks in the style of a corpus, grammaticon, etc.? In other words, contrary to what linguists seem to believe—namely, that structured resources are the future of linguistics—AI models seem to have an issue with them, and AI is the epitome of a tool related to the future.

Returning to our central question of how linguistic knowledge enters AI, as already indicated, an AI model knows only as much about linguistics and linguistic analysis as it has learned during training. Of course, this knowledge can be further enlarged during use through so-called retrieval-augmented generation (RAG) resources. However, the addition of such knowledge is highly limited and, as a rule, disappears when the connection to the sources from which it comes is disrupted.

Significantly, an AI model, especially one whose tokenizer is publicly accessible, such as ChatGPT, allows glimpses into what happens inside the model, and in many cases we can determine whether specific linguistic information is used. Thus, contrary to all expectations, inflectional paradigms, for example, are not represented in ChatGPT, even though generative morphologists who did not previously believe in paradigms have started using them. The distinction between derivation and inflection is not represented in an AI model either; thus, there are no derivational paradigms. The same holds for syntactic trees and any type of semantics-based construction or network.

An AI model does not operate with semantics in the classical sense. Semantics-like effects arise from statistical observations about the linear co-occurrence of tokens, which immediately makes every linguistic definition based on semantics useless. Classical morphemes, which always link form and meaning, provide a very good illustration of the issue at hand, and so on.

As a rule of thumb, from a computer science point of view, everything that cannot exist without meaning specification or does not co-occur linearly makes generation more complex than the derivation of a linear, form-based structure and is therefore not of interest. To illustrate, the forms of an inflectional paradigm do not compete for the same position or co-occur in neighbouring positions in a sentence, which raises the question of the point of traversing the whole paradigm for a single form. The same argument holds for words and derivational morphemes for which many scholars postulate semantics-based constructions or networks.

I can imagine that some linguists still wonder why an LLM can produce paradigms, trees, and even networks — they were learned during training, that is, they come from linguistic texts. But the AI model does not use them internally; therefore, such structures are also most prone to errors.

Taken together, all the facts mentioned here require the relationship between linguistics and AI to be reconsidered.

Of course, one could ask why it should matter to linguists how AI generates language—after all, AI can generate language fluently, while linguists cannot. In other words, AI offers a possible solution to the task of the fluent generation of human language, without restriction to a specific language or language type. Therefore, drawing on a principle familiar from mathematics, we know that if a solution brings the desired result, it is correct. This, however, does not mean that it is the only or the most efficient solution, and this is where linguists can find space for future research.

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Forthcoming text: “Frequency versus Productivity and Why AI Does Not Use the Latter”